



Swedish Competence
Centre for Satellite-Enabled
Social Science Analytics



Rymdstyrelsen
Swedish National Space Agency

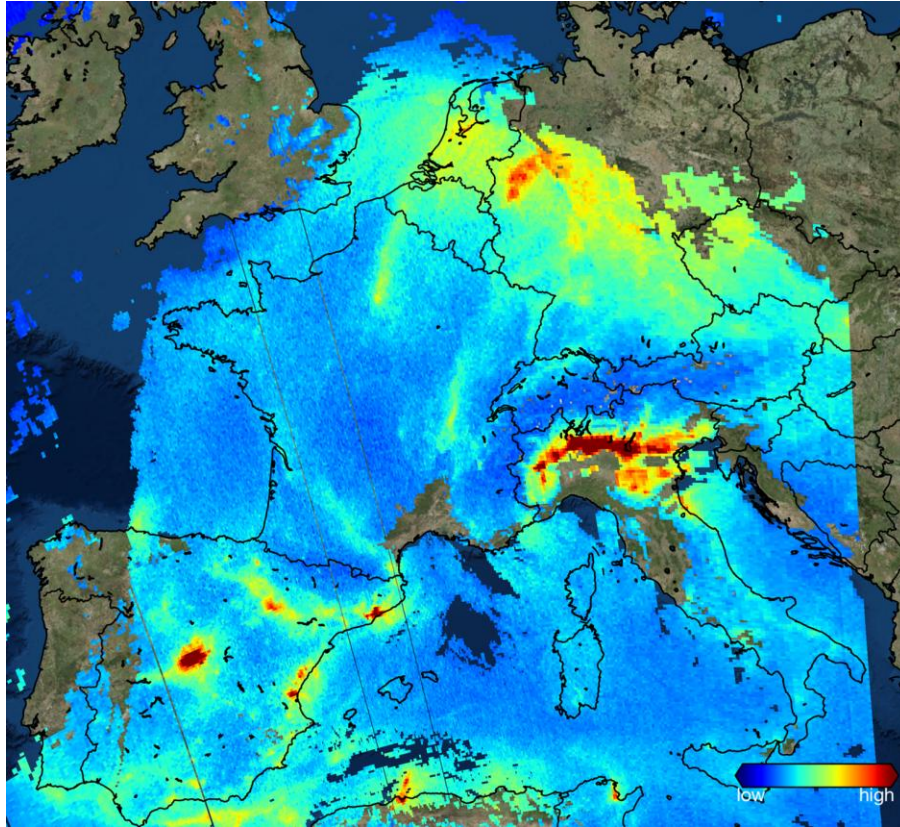
Earth Observation Fundamentals

From satellites to spatial data

Stefanos Georganos

Karlstad University · SESAC

And this?



Sentinel data (2017), processed by ESA/KNMI..

And this?



NASA Earth Observatory image by Jesse Allen

Table of contents

01

Observing Earth From Above

Tell the platforms apart: satellite, aircraft, drone — and which orbit suits which question

02

How Satellites “Sees” the Earth

How a sensor makes a pixel: energy $\rightarrow$ bands $\rightarrow$ numbers; passive vs active

03

Resolution: How Much Can We See?

Use all of four resolutions: spatial, temporal, spectral, radiometric — and their trade-offs

04

Choosing the Right EO Data

Choose a dataset and defend it: open vs commercial vs drone: coverage, revisit, cost, license

05

From Imagery to Spatial Data

Treat an image as spatial data: raster, vector, layers, CRS — and the two traps you know already

06

Where This Is Already Being Used

Place EO in your own field: what it measures well, and what it cannot measure at all

01

Observing the Earth from Above

What Earth observation is, and where the observations come from?



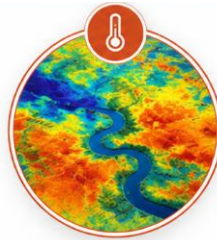
Earth observation is measurement at a distance, three choices, every time



PLATFORM

Where the instrument sits.

Ground station, drone, aircraft or satellite, that fixes how much you can cover, and how often.



SENSOR

What it is sensitive to.

Which wavelengths, how many bands, and whether it supplies its own energy or borrows the Sun's.



GEOMETRY

How it looks at the ground.

Straight down or off to the side, wide swath or narrow, and at what time of day.

Where do Earth Observations come from?

Different platforms, different perspectives, different questions.



Ground

At the surface
Continuous monitoring

~ 0 m



UAV / Drone

Low altitude
High detail

~ 100 m



Aircraft

High altitude
Broad coverage

~ 3 km

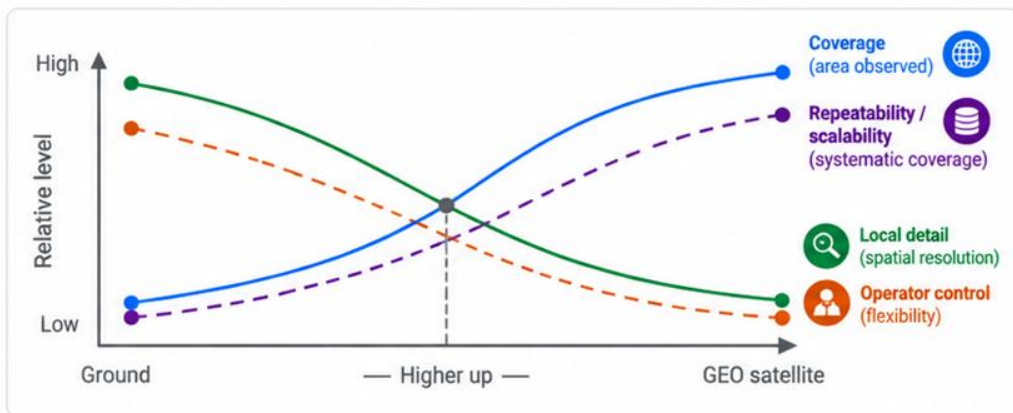


Satellite

Very high altitude
Very broad coverage

700 – 35,786 km

Platform	Height (approx.)	Detail resolved	Area covered
GEO satellite	35,786 km	1 – 3 km	a hemisphere
LEO satellite (Sentinel-2, Landsat)	700 – 800 km	10 – 30 m	a whole country
Survey aircraft	~ 3 km	decimetres	a municipality
Drone (UAV)	~ 100 m	centimetres	a few hectares
Flux tower	~ 30 m	metres	one plot
Soil moisture probe	in the ground	centimetres	one point



How do satellites observe the Earth repeatedly?

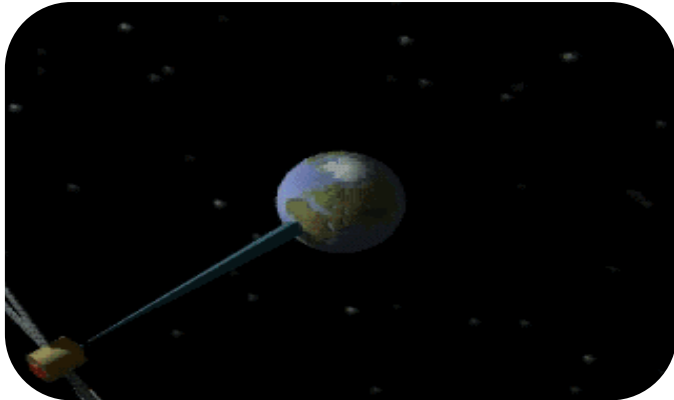
Different orbits, different ways of observing



Geostationary orbit

A fixed view of (almost) the same region

- Very high above the Earth (~ 35,786 km)
- Remains over approximately the same region
- Continuous observation
- Example: weather monitoring



Polar / near-polar orbit

A global view over time

- Travel around the Earth from pole to pole
- Earth rotates underneath the orbit
- Gradually observes different parts of the planet
- Example: land & environmental monitoring



How do satellites observe the Earth repeatedly?

Different orbits, different ways of observing



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Or use QR code

Module 01: Observing Earth From Above



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02

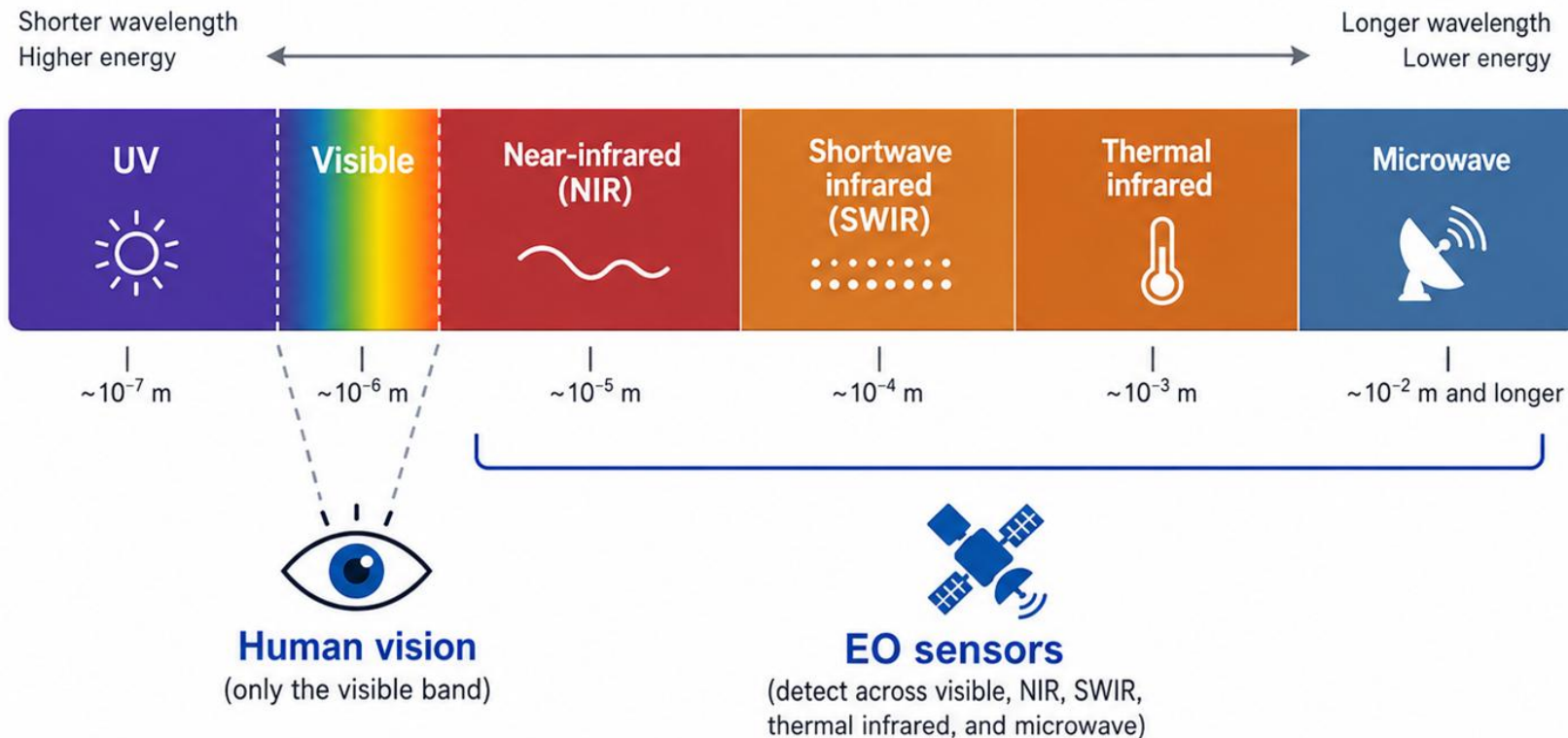
How Satellites “See” the Earth

Sensors, wavelengths, bands and pixels



See Beyond Human Vision

The Electromagnetic spectrum



Same Place, Different Information

Different wavelengths reveal different properties



The same place can look very **different at different wavelengths.**



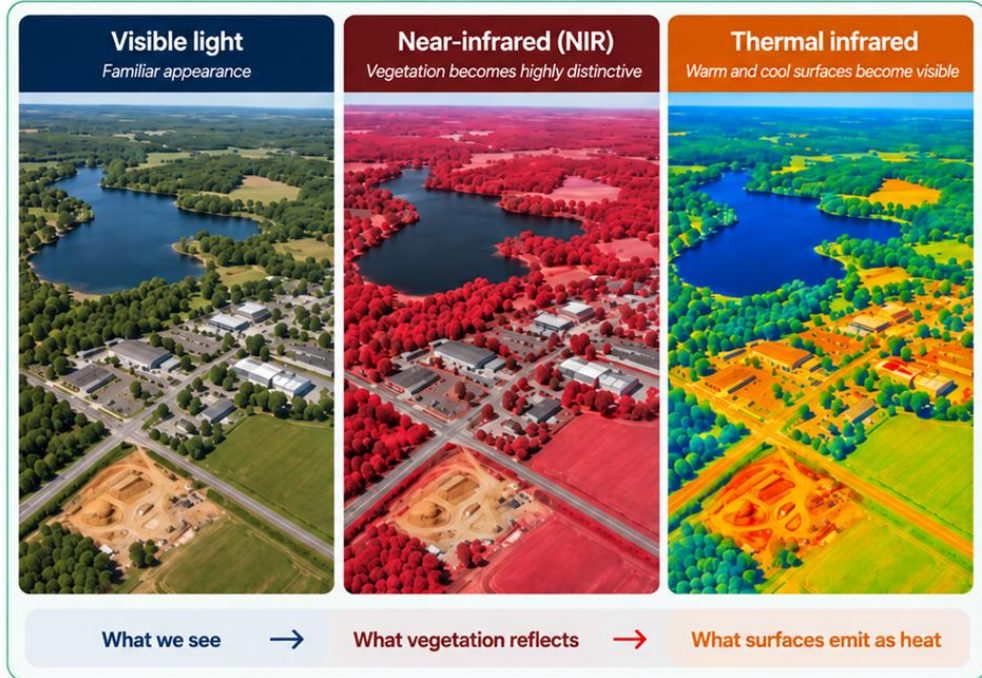
Different materials **interact with energy** in different ways.



These differences help us identify properties that are **invisible to our eyes.**



Different wavelengths provide **different information** about the same landscape.



What Happens to Energy at the Earth's Surface?

Reflection, absorption, emission and backscatter



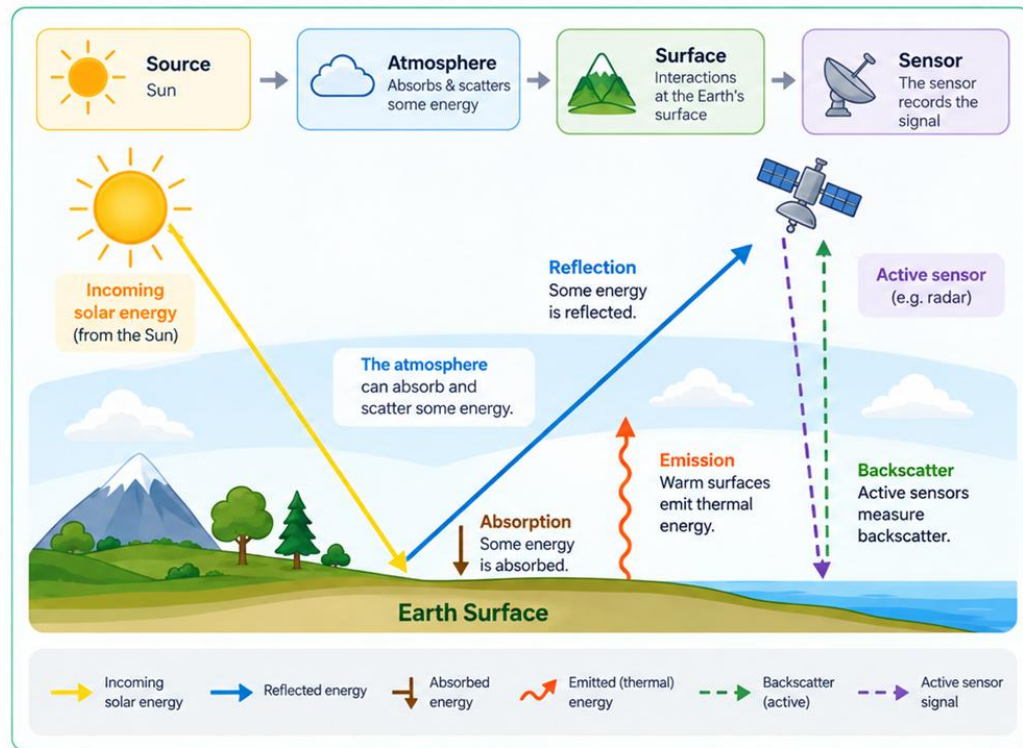
Energy interacts differently with **different surfaces**.



What reaches the sensor depends on what is **reflected, emitted, or returned**.



These interactions create **measurable differences in EO data**.



Different Surfaces, Different Responses

Why vegetation, water and built surfaces look different



Different materials have different **spectral and thermal responses**.



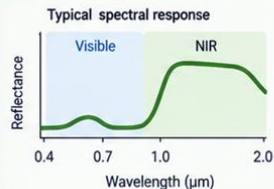
The pattern of response across wavelengths is often called a **spectral signature**.

1. Vegetation

Plants are highly distinctive



- Strong reflection in near-infrared (NIR)
- Healthy vegetation stands out clearly

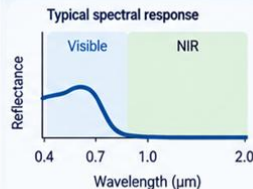


2. Water

Absorbs most radiation



- Strong absorption in NIR/SWIR
- Often appears very dark in those wavelengths

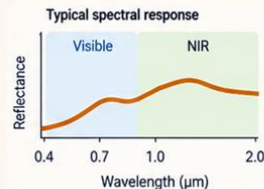


3. Built surfaces

Varied materials, mixed signals



- Different materials reflect differently
- Concrete, roofs, asphalt and soil have distinct responses

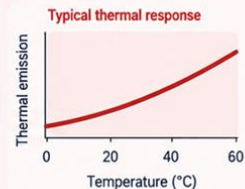


4. Warm surfaces

Emit more thermal radiation



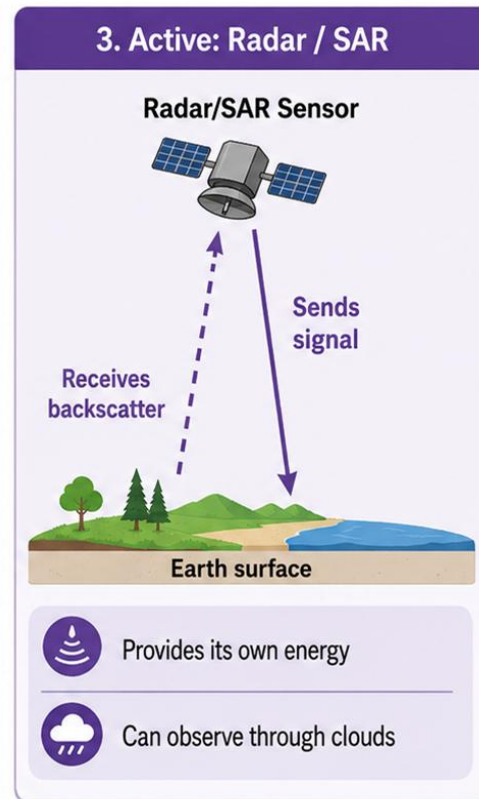
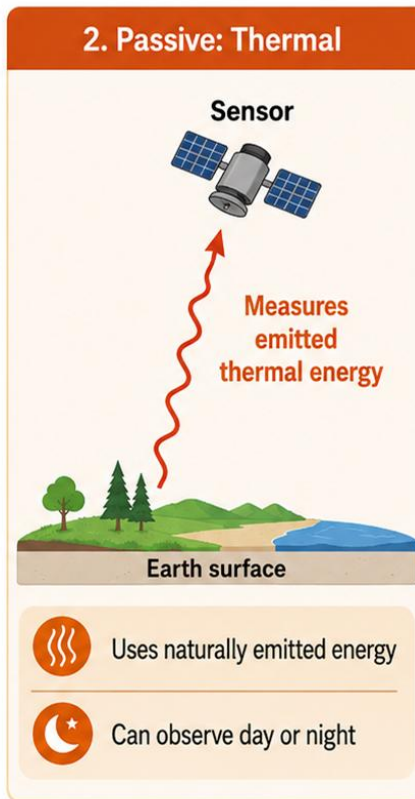
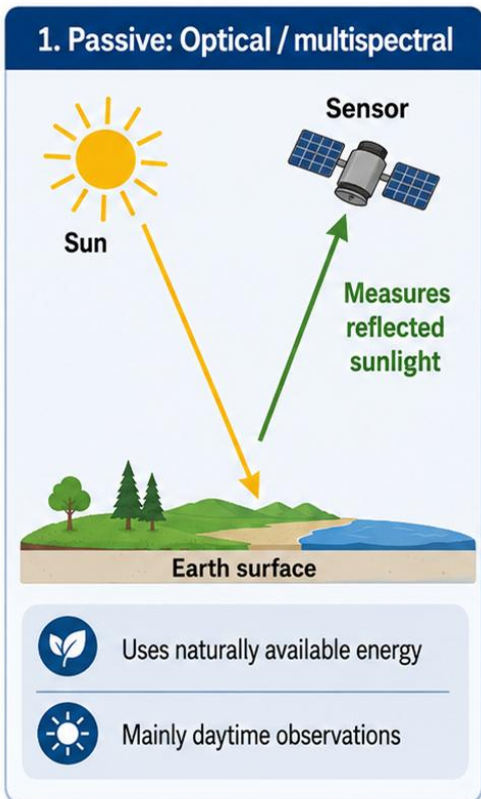
- Emit thermal infrared radiation
- Warmer surfaces emit more thermal energy



Different materials leave different **measurable “signatures”** in Earth Observation data.

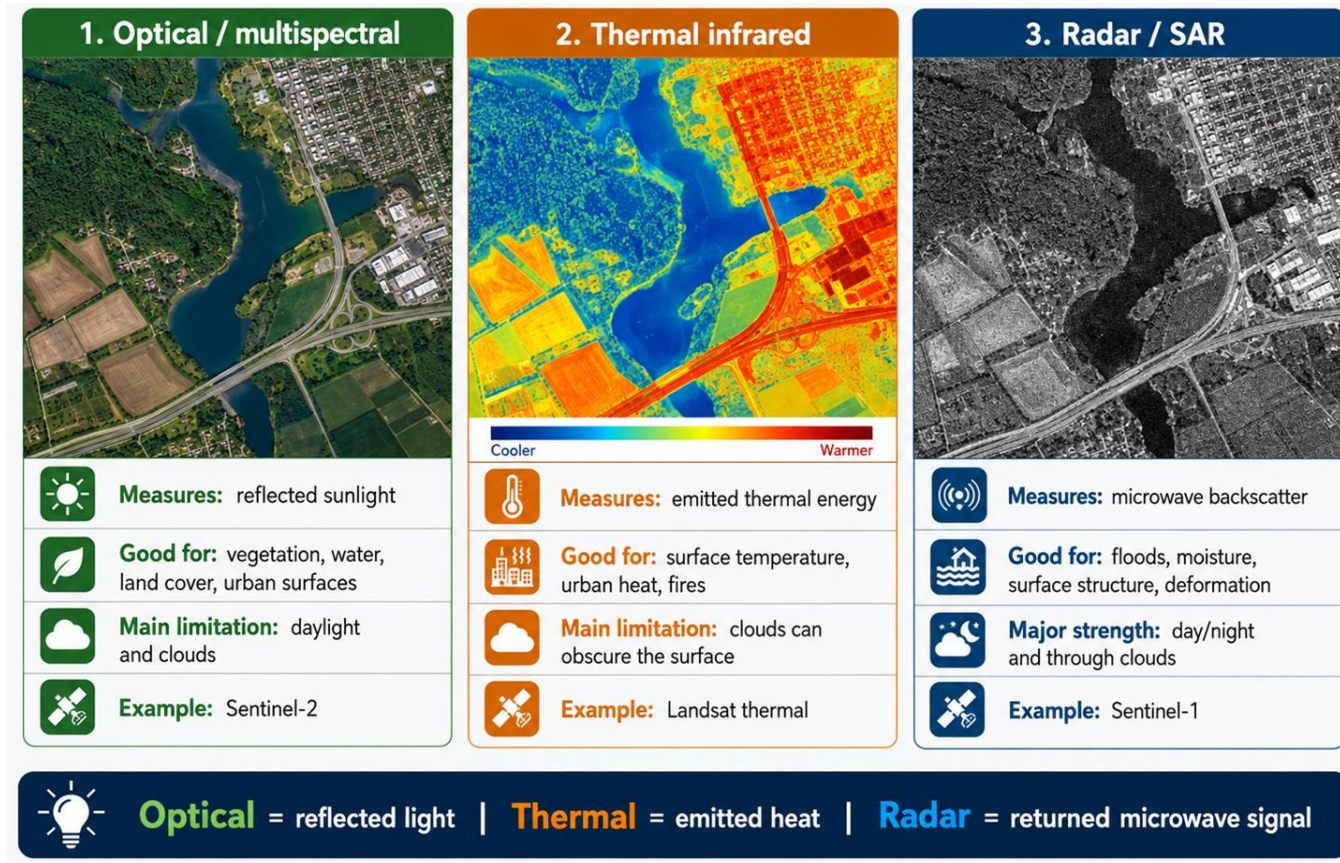
Passive vs Active Sensing

Where does the measured energy come from?



Three ways of observing the same earth

Optical, thermal and radar reveal different properties



From Signal to Pixel

An image is a grid of measurements



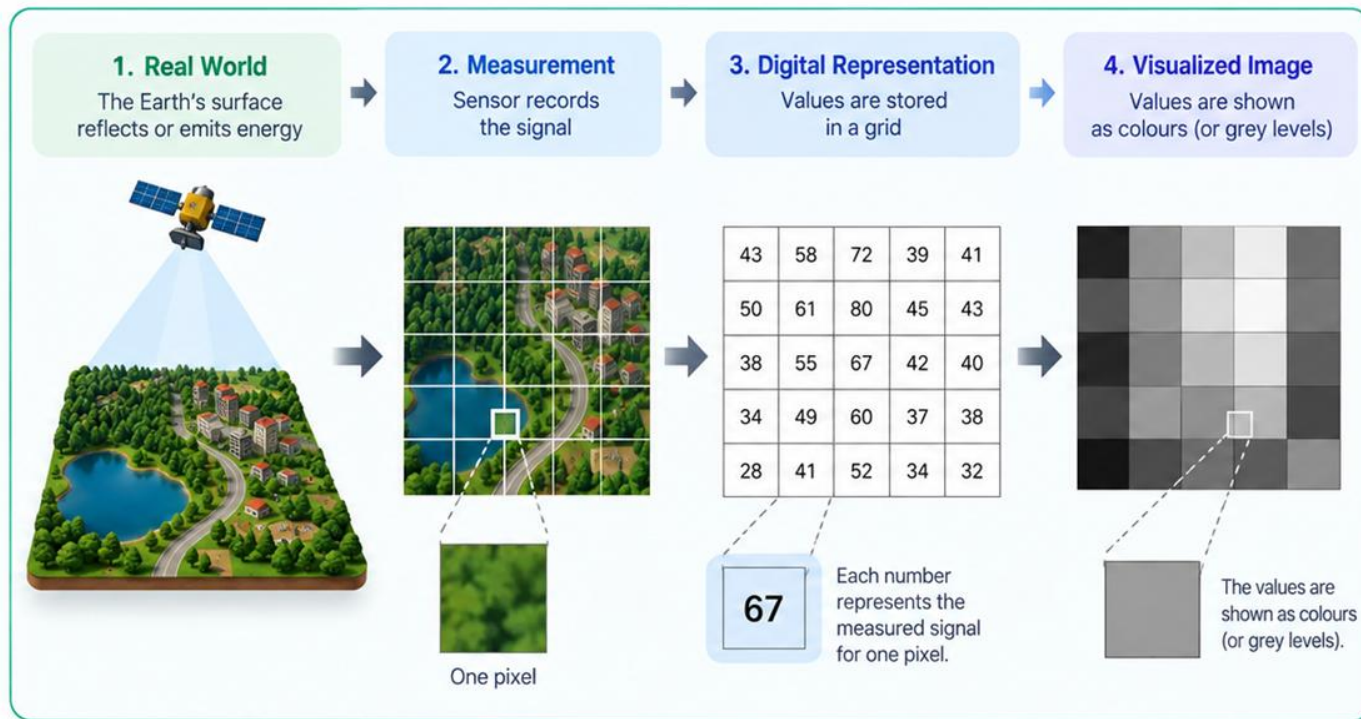
Sensors convert detected energy into **numerical values**.



EO images are organized as a **grid of pixels**.



Each pixel contains a **measurement** (e.g. reflectance, brightness, temperature).



From One Band to Many

Different wavelengths become different layers



A **band** records measurements from a particular wavelength range.



RGB imagery combines **three bands**.



Multispectral sensors record **several bands**.



Different wavelengths provide **different layers of information** about the same place.

Original Scene

The same place on Earth



Individual Spectral Bands

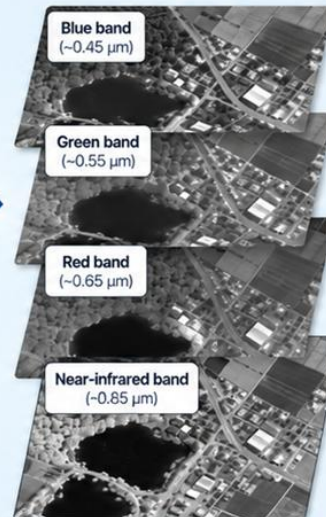
Each band shows a different part of the electromagnetic spectrum

Blue band
(~0.45 μm)

Green band
(~0.55 μm)

Red band
(~0.65 μm)

Near-infrared band
(~0.85 μm)



Red, Green, Blue

Three bands combined (natural colour)



NIR, Red, Green

Different bands reveal new information (false colour)



Predict before you look: what changes when we swap in the near-infrared band?



Contains modified Copernicus Sentinel data 2025, processed via Copernicus Data Space Ecosystem.



Module 02: How Satellites "See" the Earth



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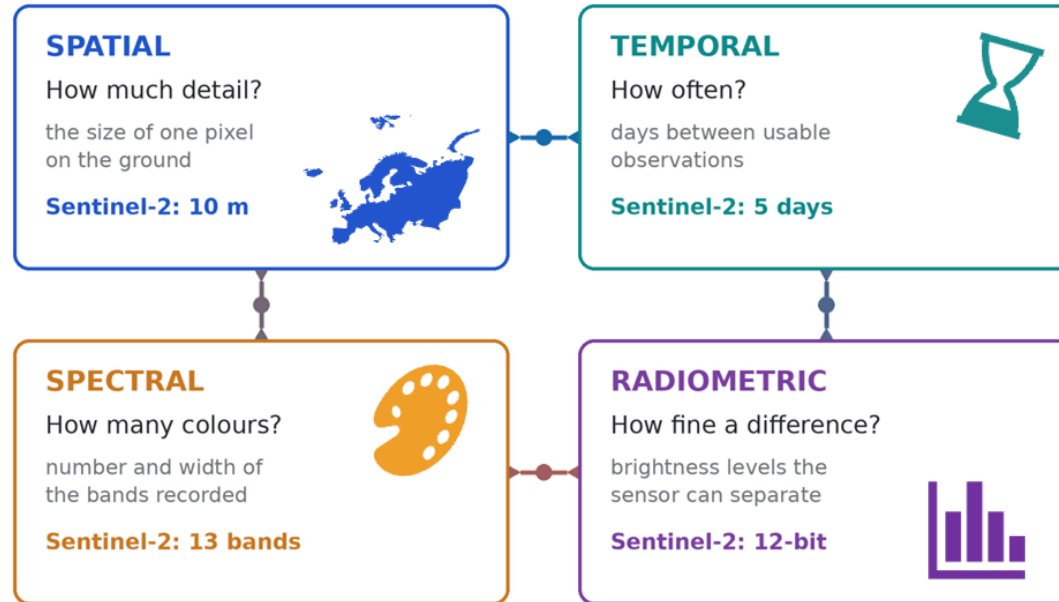
03

Resolution: How Much Can We See?

Spatial, temporal, spectral — and the fourth one nobody mentions



“Resolution” is four different questions, and they trade against each other



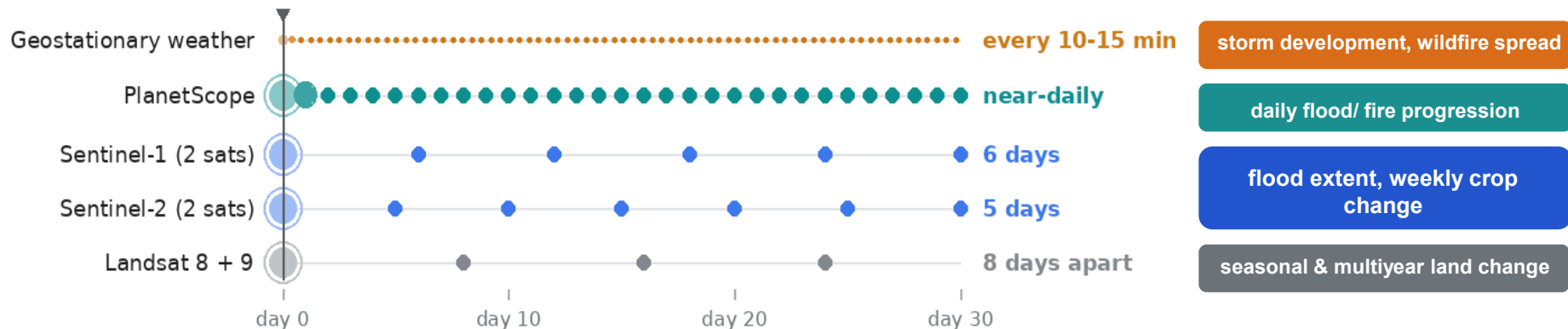
Spatial Resolution: How Much Detail?

How much ground does one pixel represent?



Temporal Resolution: How often?

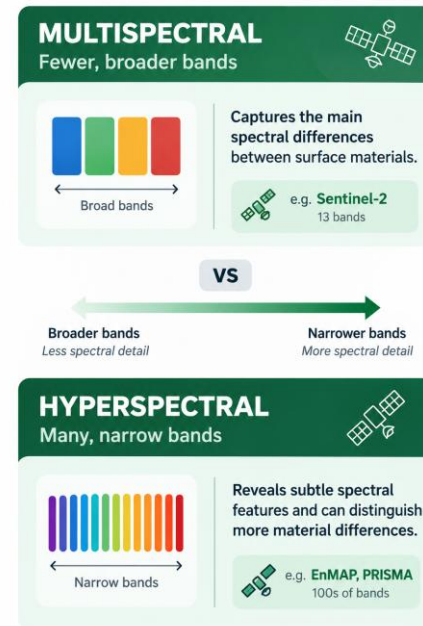
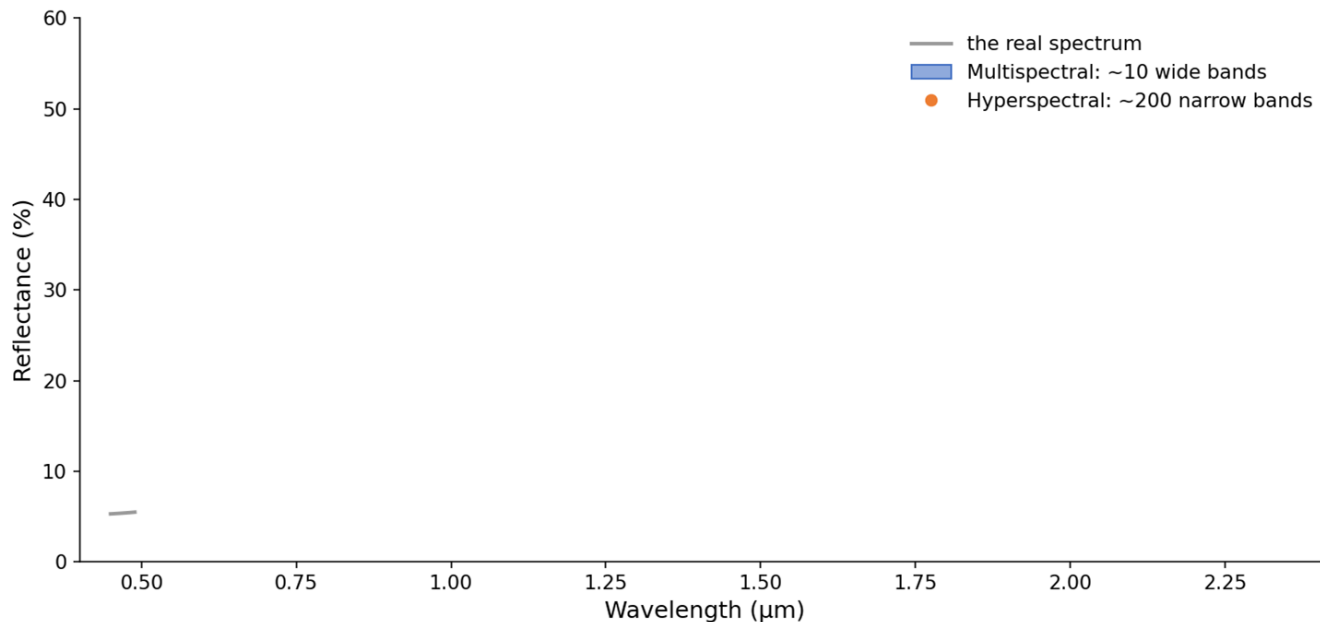
Time between repeated observations



Each dot is one scheduled overpass. Clouds remove some of the optical ones; radar keeps all of its own.

Spectral Resolution: How much wavelength information?

How finely do we divide the electromagnetic spectrum?



Radiometric Resolution: How fine a difference?

How precisely can a sensor distinguish signal intensity?



Each pixel records
a signal intensity
(brightness).



Radiometric resolution
determines how many
different intensity levels
the sensor can distinguish.



More levels allow
subtler differences
to be detected.



Radiometric resolution = the number of distinct intensity levels a sensor can detect.

More intensity levels
help us see more details about
the Earth's surface.

8 levels (3-bit)

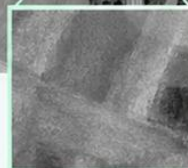
$2^3 = 8$ possible intensity values



More pixelated
Fewer intensity levels

256 levels (8-bit)

$2^8 = 256$ possible intensity values



More detail
More intensity levels

Intensity levels (grayscale)

8 levels (3-bit)



Values: 0 1 2 3 4 6 7

256 levels (8-bit)



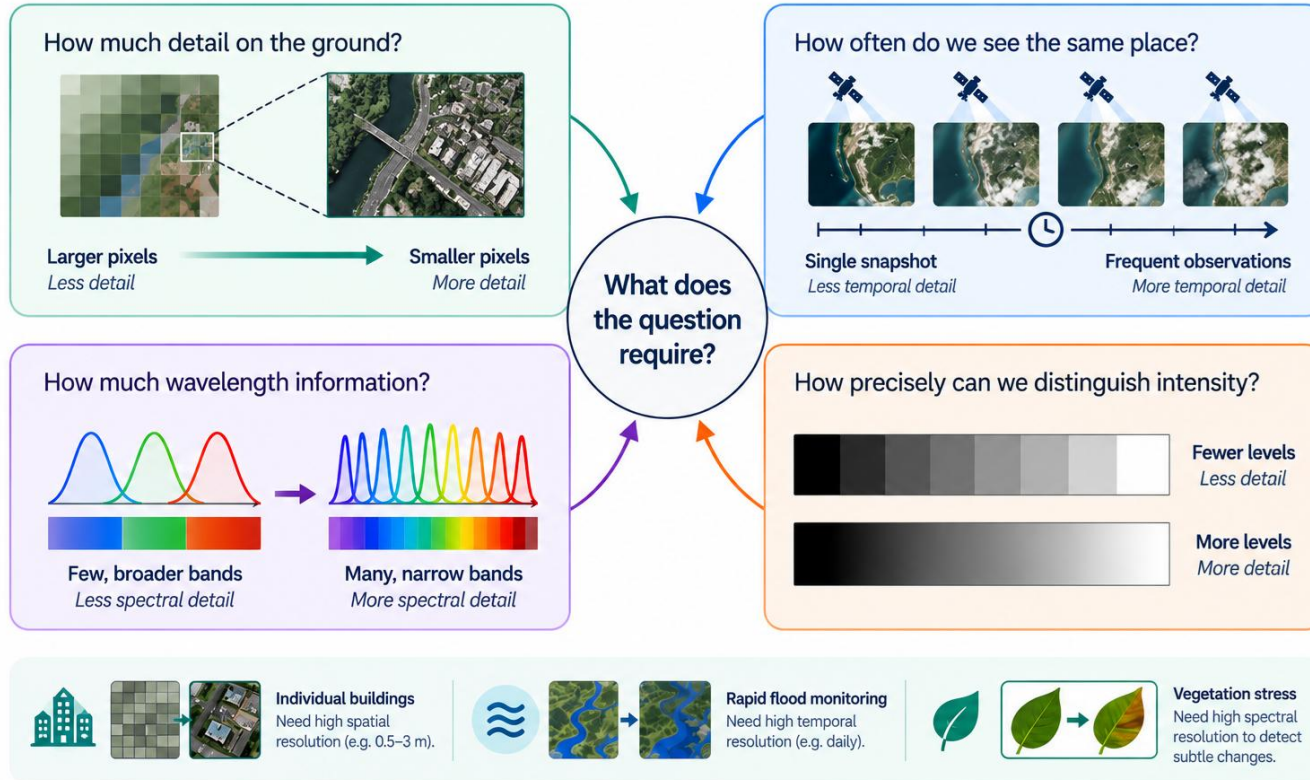
Values: 0 ... 255



Higher radiometric
resolution captures
finer variations
in signal intensity.

Resolution Trade-offs: What does the question require?

Higher resolution is not always better



Choose the resolution that fits your question – it's about the right balance, not the highest possible.

Module 03: Resolution



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04

Choosing the Right Earth Observation Data

Open, commercial and drone-based observations?

Start with the Open EO Ecosystem

Sentinel, Landsat and freely available observations

Dataset / source	What it provides	Access
 Copernicus Sentinels	Optical, radar, ocean, land & atmosphere observations	Free & open
 Landsat	Optical + thermal imagery, long historical archive	Free & open
 VIIRS night lights	Global night-time light observations	Free & open
 Copernicus DEM	Global elevation data	Free access
 GHSL	Built-up, settlement & population layers	Open
 OpenStreetMap	Roads, buildings, facilities & other mapped features	Open



Many global EO datasets are available freely and openly.



Open missions provide consistent, repeated observations over large areas.



Satellite imagery is often combined with other open geospatial datasets.



Takeaway: For many questions, open data is the best place to start.

The open EO Workhorses



Sentinel-2

Optical detail for land and water



MEASURES
reflected light
13 bands



RESOLUTION
10 / 20 / 60 m



REVISIT
5 days



SWATH
290 km



RECORD SINCE
2015



SEES THROUGH CLOUD
no



FREE
Open and accessible data



Sentinel-1

All-weather radar imaging



MEASURES
radar backscatter
C-band (SAR)



RESOLUTION
5 × 20 m



REVISIT
6 days



SWATH
250 km



RECORD SINCE
2014



SEES THROUGH CLOUD
yes



FREE
Open and accessible data



Landsat 8/9

A long record of our changing planet



MEASURES
light + heat
11 bands



RESOLUTION
30 / 15 / 100 m



REVISIT
8 days



SWATH
185 km



RECORD SINCE
1972



SEES THROUGH CLOUD
no



FREE
Open and accessible data

Commercial Satellite Imagery

More detail, more timing control but with cost and licensing



What you get

- WorldView Legion: six satellites, 0.29 m panchromatic, up to 15 passes a day.
- Planet: 3 m, near-daily, everywhere.
- Tasking — you specify the date and the place.



When it is worth it

- You must resolve individual buildings, roofs or damage.
- You need imagery of a specific day that no open sensor covered.
- You need it faster than the open archive delivers.



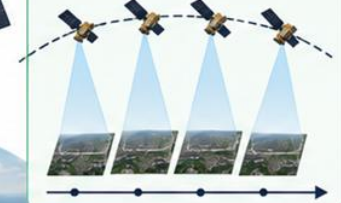
The trap: the licence, not the price

- Data can be expensive.
- Usage rights may be restricted (commercial licence).
- Always check the terms and intended use.

Sub-metre detail



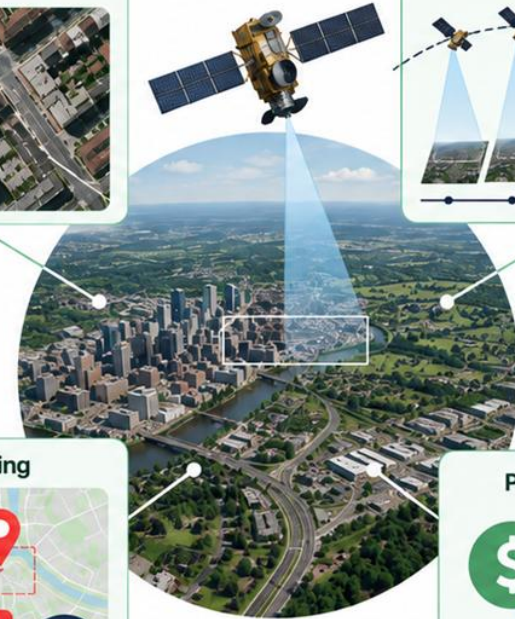
Frequent observations



Tasking



Paid / licensed



Drone-based Remote Sensing

Maximum local detail and acquisition control



What you get

- Centimetre-level spatial detail
- High control over **where and when** data are **collected**
- Flexible sensors and flight plans
- Detailed products such as **orthomosaics, surface models and 3D data**



What it requires

- Limited area coverage compared with satellites
- **Fieldwork, flight planning and processing**
- Weather, regulations and permissions can constrain acquisition
- Large areas can become time-consuming and data-intensive

Centimetre detail



Different sensors



Fly when needed

- You choose the area and timing.
- Flexible flight plans.



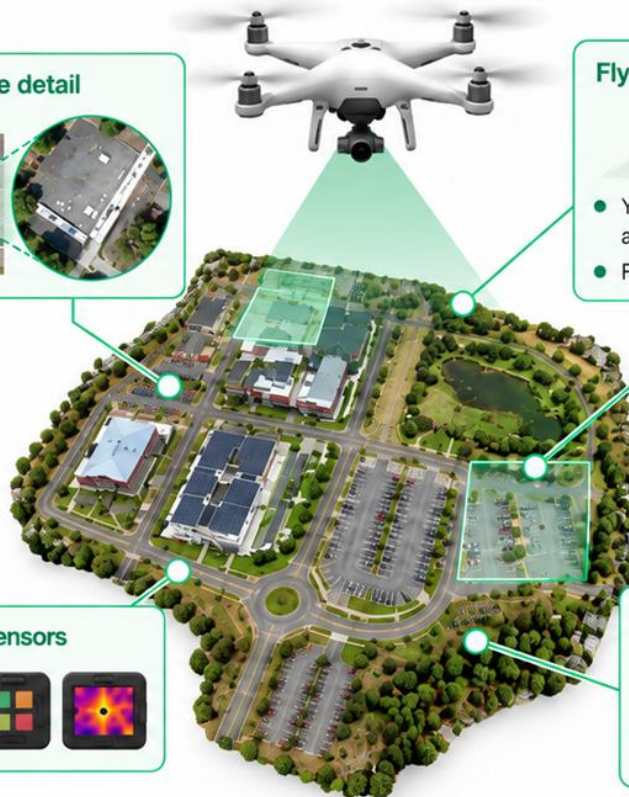
Detailed products

- Orthomosaics, surface models and 3D data.



Small-area coverage

- Limited area compared with satellites.
- Ideal for targeted sites of interest.



The Practical Trade-space

Coverage, detail, frequency, control and affordability



Open satellites

Broad, repeatable and inexpensive coverage



Commercial satellites

High detail and stronger acquisition flexibility



Drones

Maximum local detail and control, but limited coverage and field effort

	 Open Satellite	 Commercial VHR	 Drone
 Area coverage	Very high	High	Low
 Spatial detail	Moderate	Very high	Extremely high
 Observation frequency	Regular / repeated	Frequent / taskable	On demand*
 Acquisition control	Low	High	Very high
 Affordability	Very high	Low	Medium
 Operational effort	Low	Low	High

*Subject to weather, permissions, logistics and field access

Pick the Question First, the Sensor Last

Choose data from the problem, not the other way around



The wrong way

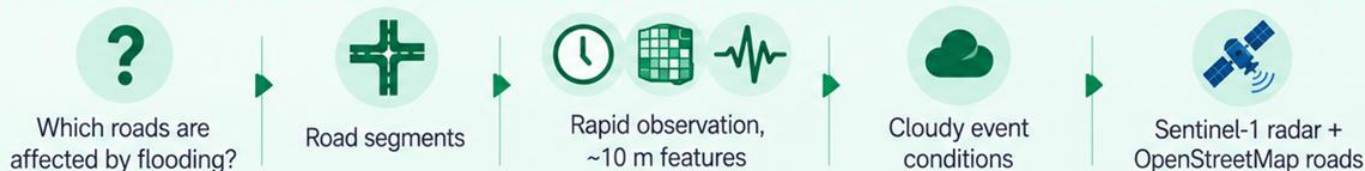
"I have Sentinel-2 — what can I do with it?"



The better way



Worked example: Flooding



Module 04: Choosing the Right EO Data



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05



From Satellite Imagery to Spatial Data

Connecting Earth observation and GIS — and two traps you already know

An EO image has a location

From image to georeferenced raster layer



EO imagery is usually
georeferenced



Each **pixel** corresponds
to a specific location
on earth.



This allows imagery to
align with maps and
other spatial datasets.



A **coordinate reference
system** tells GIS how
those locations are
represented.



GIS Represents the World in Two Main Ways

Raster data and vector features



Raster = values across space (Eg: satellite imagery, temperature, elevation, land cover)



Vector = identifiable geographic features (Eg: point for trees, line for road, polygon for buildings/boundary)

Raster

Grid of pixels



Vector

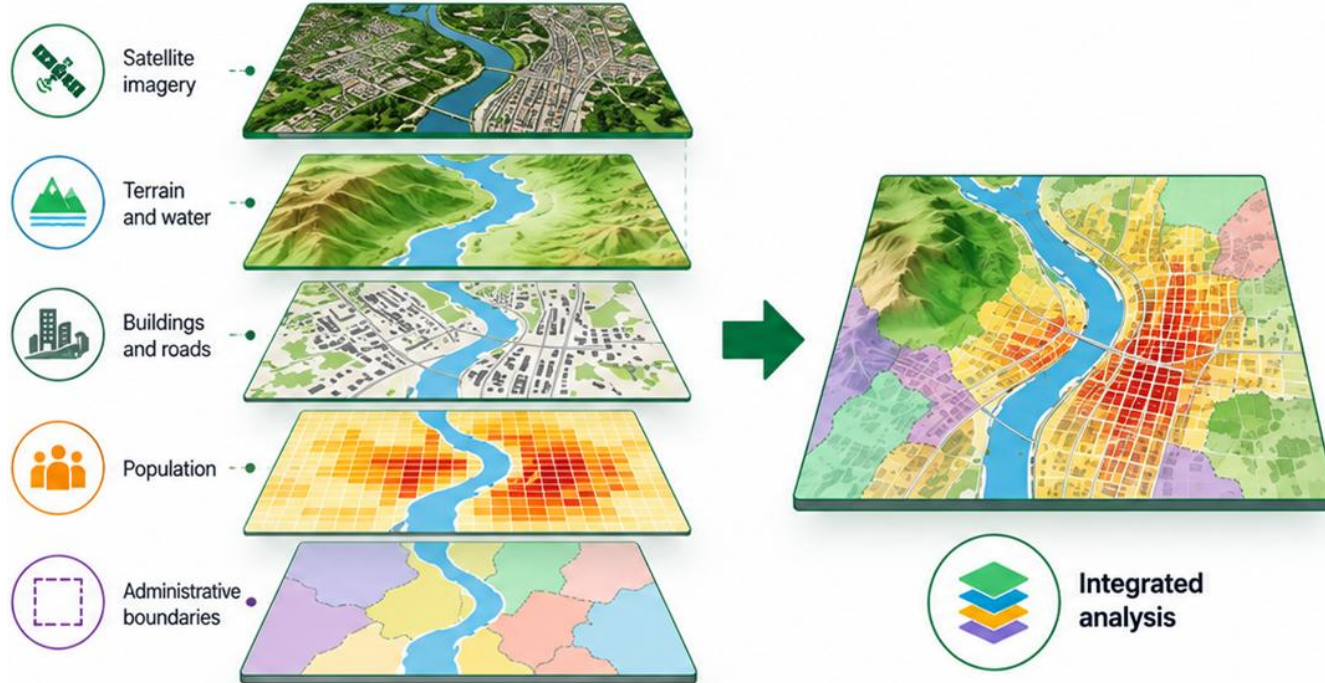
Points, lines, polygons



Raster describes values across space; vector represent geographic features.

Adding Context: Stack the Layers

EO rarely answers the whole question alone



EO provides **one layer** of information about the Earth.



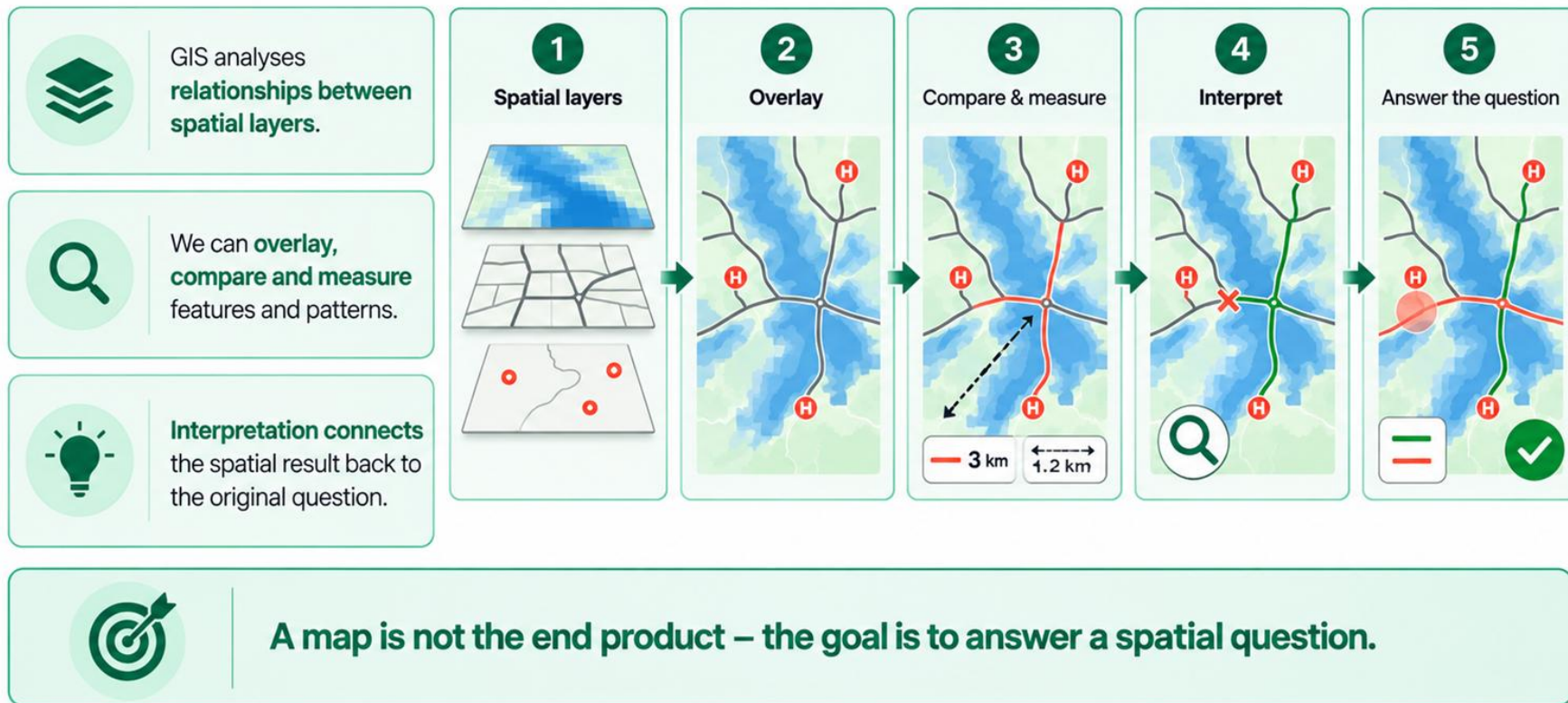
Other spatial layers **add context**: infrastructure, people, terrain, boundaries.



GIS **combines** these layers to support analysis and interpretation.

From Layers to Spatial Questions

Overlay, compare, measure and interpret



Module 05: From Satellite Imagery to Spatial Data



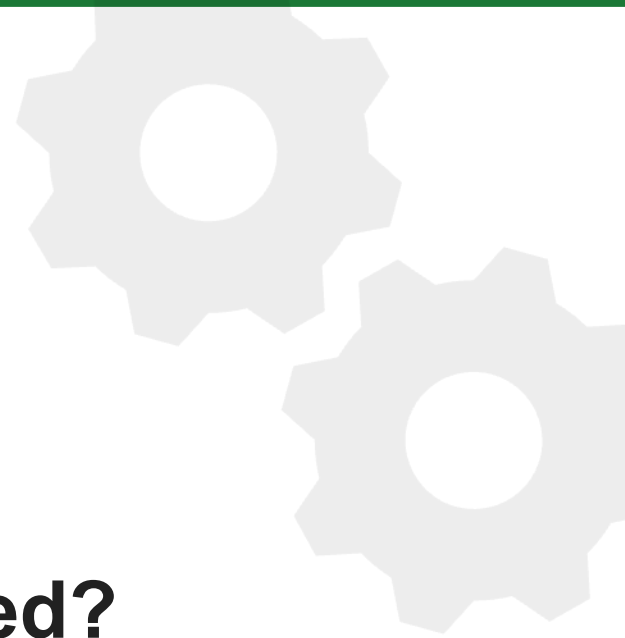
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06

Where This Is Already Being Used?

Breadth across disciplines — and where the evidence is thin



One Technology, Many Disciplines

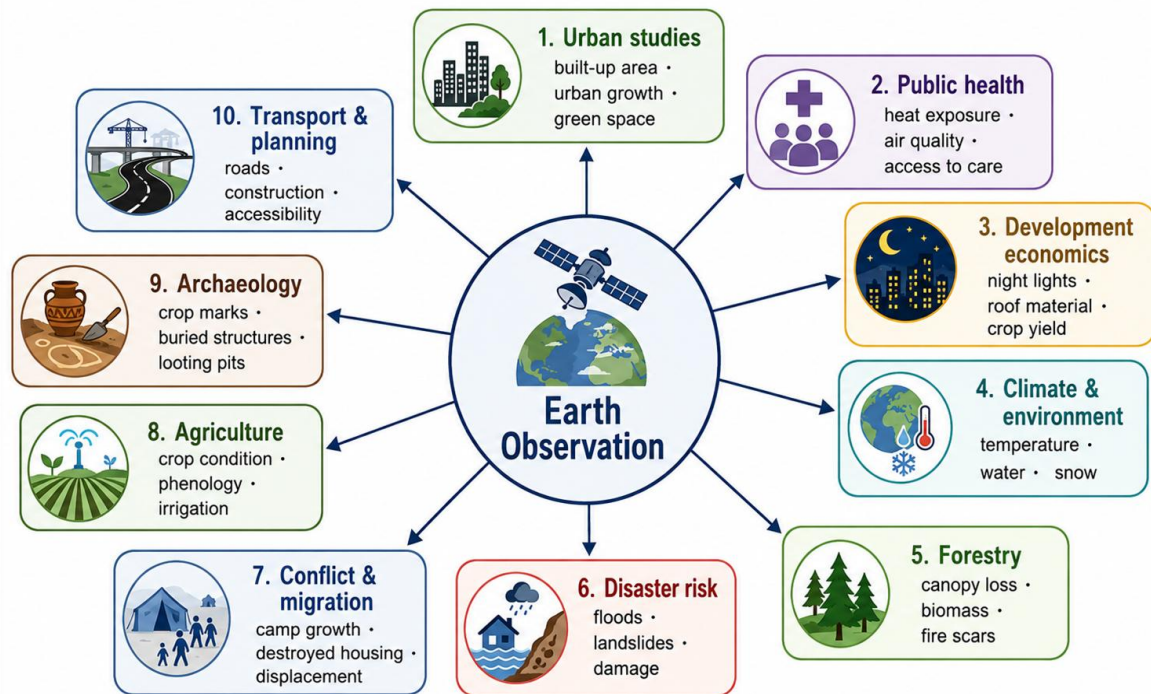
The same observations support very different questions



EO provides **repeatable observations** across large areas and through time.



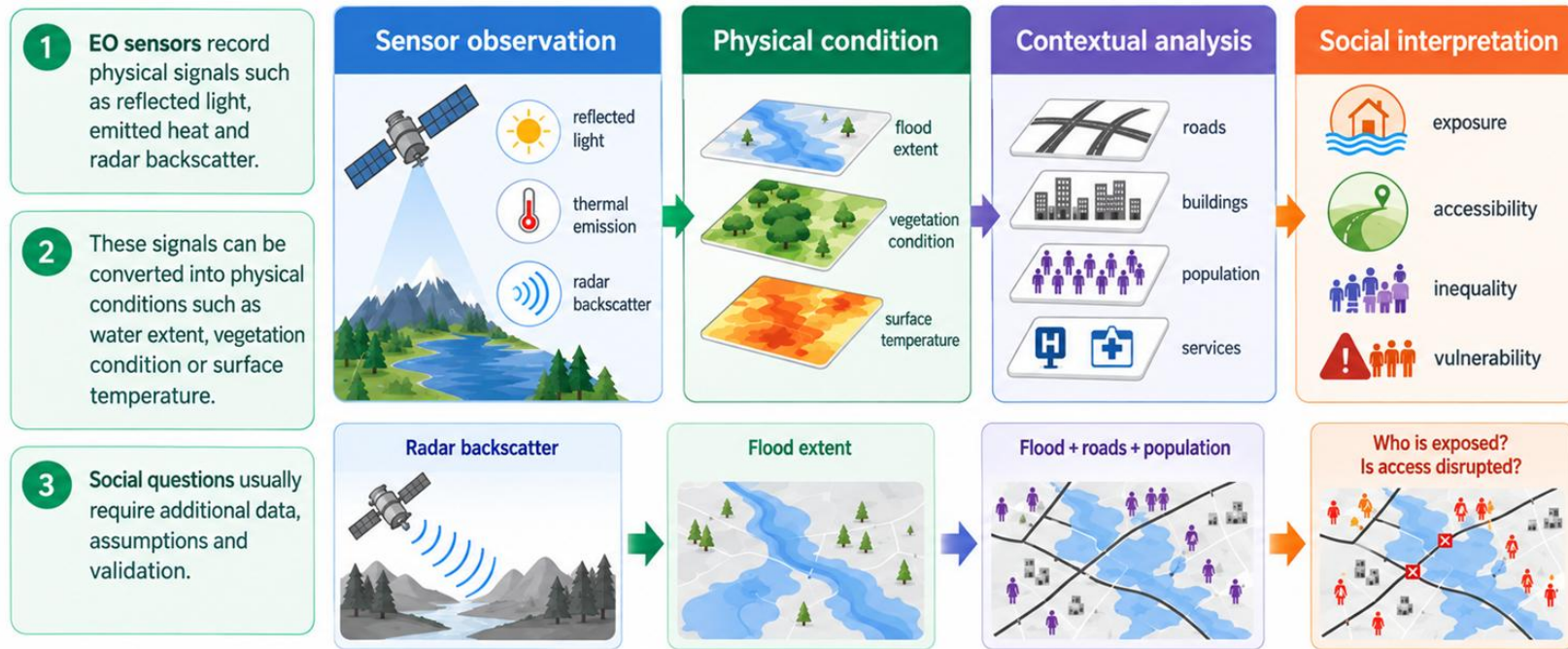
Different disciplines turn those observations into **different questions and applications**.



The same Earth Observation answer questions across very different fields.

From Direct Measurements to Derived Applications

Physical observation → contextual analysis → social inference



The further the question moves from physical observation, the more additional evidence matters.

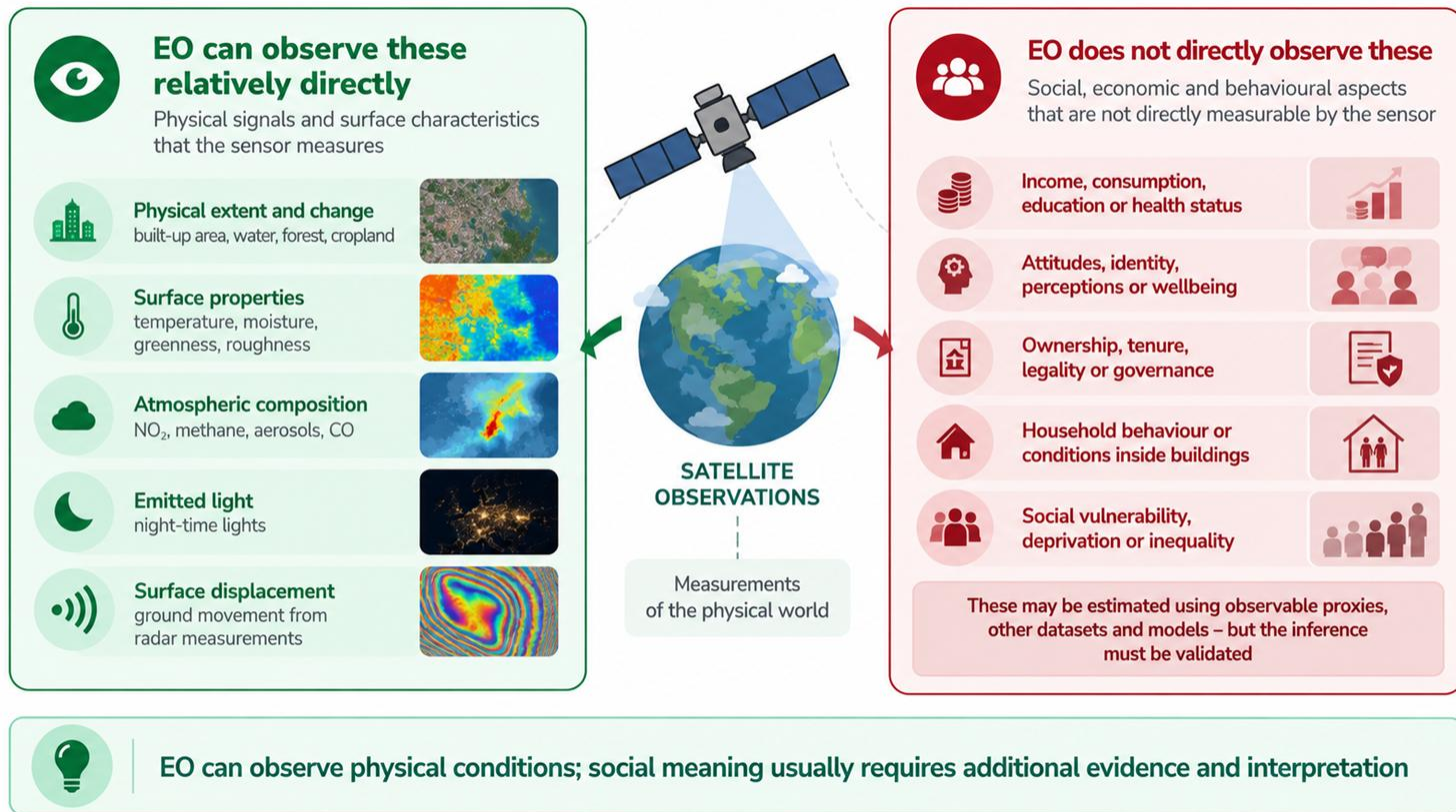
You Often Do Not Need to Start from Pixels

Many useful EO products and services already exist



Know the Boundary

What EO observes directly and what must be inferred



Module 06: Where This is Already Being Used



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Waiting for participants

Six things, one sentence each...

- 1 **Orbit height and platform fix** what you can ask before you choose anything else.
- 2 **A sensor turns energy into bands**; radar makes its own, which is why it beats cloud.
- 3 **“Resolution” is four questions** — and finer is not better, it is different.
- 4 **Start with open data**; pay only when you can name what the money buys.
- 5 **An image is a table with coordinates** — so MAUP and the ecological fallacy apply.
- 6 **EO measures the visible and infers the rest**. The inference is your job, and your skill.

The rest of this training builds directly on the last 85 minutes



TODAY
13:00

Contextualizing EO through visual interpretation

Lina Eklund — why context matters, what automated products miss, and how to read an image properly in Google Earth Pro.



TOMORROW
09:00

EO for urban and sustainable development

Growth, urban form, exposure, inequality and SDG monitoring — the applications section of this morning, at full length.



TOMORROW
10:00

Hands-on: from flood mapping to accessibility

With Amalia Chantziara. Sentinel-1 and Sentinel-2 → flood extent → roads and hospitals → who loses access. An end-to-end flood chain.



Full beginner
EO analysis series



AFTER THAT

Society from Space Challenge

The hackathon. This training is the recommended preparation for it.



Hackathon



Swedish Competence
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Social Science Analytics



Rymdstyrelsen
Swedish National Space Agency

Thank you! 😊

From fundamentals to applications — there is more to explore.

Stefanos Georganos

Karlstad University · SESAC